

Stylecart: E-Commerce App with customization options

By

Ankshika Ghosh

(B.Tech CSE 2nd Year, Enrollment No.:12023002026008)

Anshu Kumar

(B.Tech CSE 2nd Year, Enrollment No.:12023002001004)

Supratim Modak

(B.Tech CSE 2nd Year, Enrollment No.: 12023002001020)

Under the Supervision of

Prof . Debajyoti Chatterjee

Dept. of Computer Science & Engineering

University of Engineering & Management, Jaipur

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Introduction

This project focuses on building a user-friendly e-commerce mobile app using Flutter and Firebase. It offers all the core shopping features like browsing products, managing orders, and secure checkout. Beyond the basics, we added customizable product options where users can select their preferred color or add prints before ordering. A simple AI chatbot is also included to help with common queries. The goal is to deliver a smooth and personalized shopping experience with real-time support and reliable backend integration.

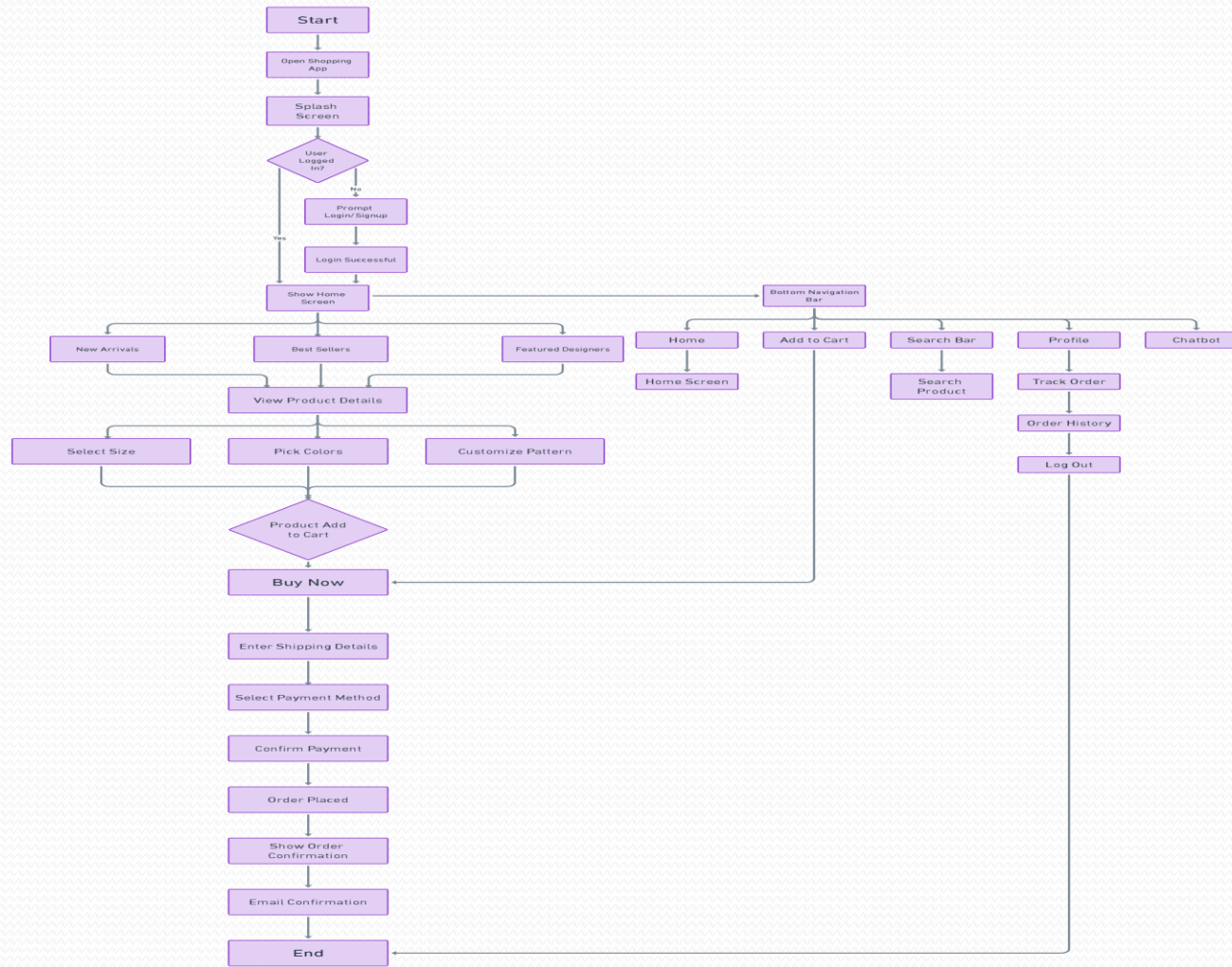
Literature review

Feature	Existing Apps (Amazon, Myntra)	Our App (StyleCart)
Product Customization	Limited to size & color options offered by seller	Full customization: choose preferred color & custom prints
AI Support	Chatbot for product search or order help (Amazon's Rufus etc.)	Simple AI chatbot for answering general queries
Order Confirmation & Tracking	Email & in-app notifications	Automated email + push notifications for every order update
Login Options	Email, Google, Facebook login	Email/Phone + Google, Apple, Facebook (OTP/password)
Wishlist & Reordering	standard wishlist & reorder options	Enhanced reorder + smart wishlist features
Payment & Invoices	UPI, Cards, Wallets, COD (invoices via email)	All methods + auto invoice generation
Personalization	Generic suggestions, no design changes	Full user-driven color/print customization

Objectives

- ◆ **Seamless Shopping Experience:** Design a smooth, user-friendly shopping app where users can browse, choose, and purchase fashion products with ease.
- ◆ **Product Customization:** Enable users to select preferred colors and upload custom prints for items like dresses and t-shirts.
- ◆ **AI Chatbot Support:** Include a basic AI chatbot to help users with general queries and improve the support experience.
- ◆ **Flexible Login Options:** Support Google, Apple, Facebook, and phone/email login with OTP or password.
- ◆ **Secure Payments:** Provide multiple secure payment modes—UPI, cards, wallets, and cash on delivery.
- ◆ **Wishlist & Reviews:** Let users save favorites and post reviews with photos or videos.
- ◆ **Order Management:** Offer order tracking, order history, and one-tap reordering.
- ◆ **Real-Time Notifications:** Send push alerts for order updates, confirmations, and offers
- ◆ **Firebase Integration:** Used Firebase for real-time data, authentication, and cloud storage.

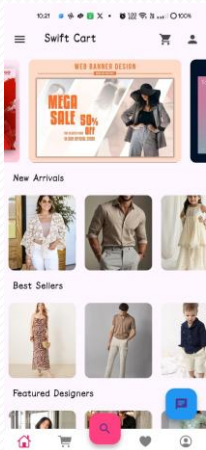
Proposed Model



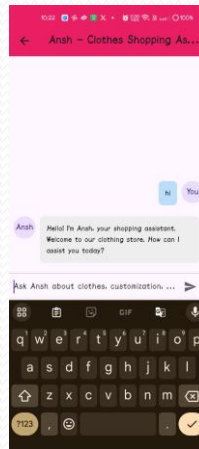
Experimental Set-up

The eCommerce application was developed using Flutter, incorporating essential packages such as cupertino_icons, gap for widget spacing, and carousel_slider for image carousels. Firebase was used for authentication, real-time database (Firestore), and cloud storage. Shared Preferences and Flutter Secure Storage managed local and sensitive data. Google APIs enabled location and time zone-based features, while EmailJS facilitated in-app email communication. A chatbot provided real-time customer support. Additionally, future implementation of 3D models and AR features was considered to enable virtual try-on and enhance user interaction within the shopping experience.

Result



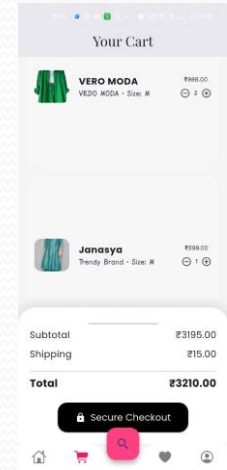
1.1 Home Page



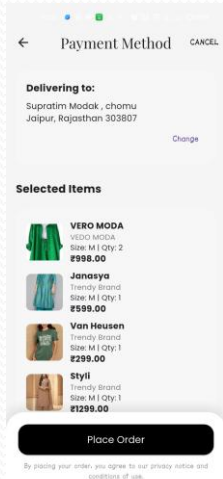
1.2 ChatBot



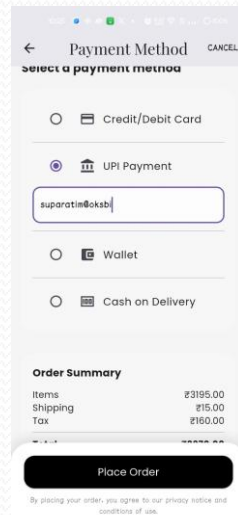
1.3 Product Page



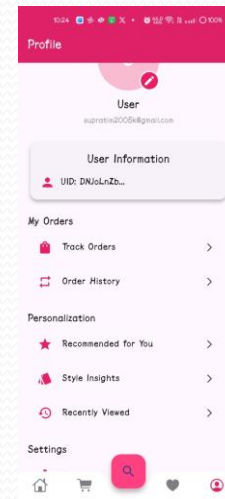
1.4 Cart Page



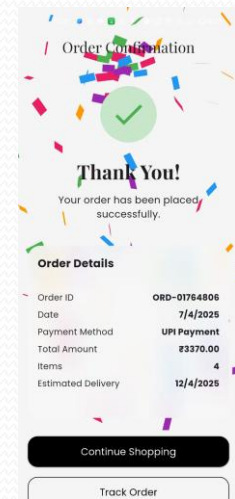
1.5 Payment Page



1.6 Payment method Page



1.7 Profile page



1.8 Order confirmation page

Limitation

- ❑ **Limited Global Scalability** : The app is designed with a regional focus and may not support international deliveries or currencies, limiting its reach for global users.
- ❑ **Customization Processing Time** : Customized items with user-chosen colors or prints may take longer to process and deliver compared to standard products.
- ❑ **Basic Chatbot Capabilities** : The integrated AI chatbot handles only general queries and lacks advanced natural language understanding or multilingual support.
- ❑ **Dependency on Internet Connectivity**: Most core features like product browsing, customization, and order placement require a stable internet connection. Limited functionality is available offline.
- ❑ **Device Compatibility for Customization Features**: Color selection and print upload features may not perform optimally on very low-end devices, affecting user experience in such cases.
- ❑ **Security Maintenance** : Though secure payment gateways are used, constant updates and monitoring are essential to safeguard user data and transactions from emerging threats.

Conclusions

SwiftCart transforms online shopping by blending smart technology, user personalization, and a clean interface to deliver a seamless and interactive fashion retail experience. The app addresses common challenges in the fashion e-commerce space—like limited customization, delayed support, and static product listings—by offering dynamic, user-driven features.

With real-time inventory updates, customizable color and print options, and a built-in AI chatbot for quick query resolution, SwiftCart is engineered for flexibility and convenience. These features don't just enhance usability—they help shape a shopping journey that reflects each user's unique taste and expectations.

Built on a modular Flutter-Firebase stack, the app ensures fast scalability, secure transactions, and cross-platform compatibility, while allowing continuous updates to meet evolving market trends and fashion preferences.

SwiftCart is more than a shopping app—it's a step toward the future of personalized fashion. It empowers users to design, discover, and shop with ease, bringing their style ideas to life right from their smartphones.

SwiftCart — Your Style, Your Way!

Future Scope

- **Augmented Reality (AR) Integration:** We plan to integrate AR-based virtual try-on features, allowing users to preview how clothes will look on them in real-time using their device camera.
- **Smart Recommendation System:** Future versions may include AI-powered recommendation engines that adapt to user behavior, style preferences, and previous purchases to suggest the most relevant fashion items.
- **Personalized Outfit Bundling:** The app can be upgraded to suggest complete outfit combinations (tops, bottoms, accessories) based on occasion, weather, or current trends.
- **Multilingual & Regional Support:** To make the app more accessible, we aim to add multiple language options and regional fashion suggestions tailored to user location and culture.
- **Virtual Size Assistant:** An AI-driven sizing assistant may be introduced to recommend the perfect fit based on body measurements, brand sizing differences, and user feedback.
- **Smart Order Fulfillment & Tracking:** Future updates will focus on optimizing logistics through smart delivery prediction, return management, and live tracking of shipments.
- **Seller Integration Platform :** Allowing local fashion brands or independent creators to list and customize products directly through a seller dashboard could expand the product variety.

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Thank You!

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